



Material flow analysis and life cycle assessment of food waste bioconversion by black soldier fly larvae (*Hermetia illucens* L.)

Hanwen Guo^a, Chengliang Jiang^b, Zhijian Zhang^{b,c}, Wenjing Lu^{a,*}, Hongtao Wang^a

^a School of Environment, Tsinghua University, 100084 Beijing, China

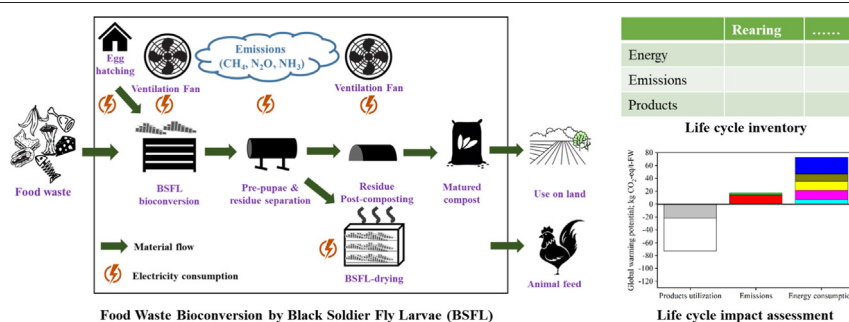
^b College of Environmental and Resource Sciences, Zhejiang University, 310058 Hangzhou, China

^c Hangzhou Gusheng Biotechnology Co., Ltd, 311108 Hangzhou, China

HIGHLIGHTS

- Life cycle inventory of a food waste bioconversion plant of BSFL was provided.
- The highest energy consumption occurred in the BSFL pre-pupae drying process.
- Utilizing fresh instead of dried BSFL pre-pupae could decrease the impacts.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 4 April 2020

Received in revised form 10 August 2020

Accepted 10 August 2020

Available online 11 August 2020

Editor: Yifeng Zhang

Keywords:

Food waste

Black soldier fly

Material flow analysis

Life cycle inventory

Life cycle assessment

ABSTRACT

This study provided a systematic analysis on material flow and environmental impacts of a food waste (FW) bioconversion plant using black soldier fly larvae (BSFL), with a daily capacity of 15 tons of FW (wet weight). Food waste feed (FWF) used for BSFL bioconversion consisted of 80% FW (collected from households, restaurants, and canteens) and 20% rice hull powder. Material flow analysis conducted on a dry weight basis showed that 6% of FWF was transformed into BSFL pre-pupae, 51% was stored in matured compost, and 43% was emitted to the air. Emissions of high environmental concern such as methane, nitrous oxide and ammonia (NH₃) were sampled and quantified by laboratory analysis. The life cycle assessment revealed that the overall impact was 17.36 kg CO₂-eq/t FW for global warming potential, 5.54 kg SO₂-eq/t FW for acidification, 24.05 mol N-eq/t FW for terrestrial eutrophication, 0.54 kg N-eq NH₃/t FW for marine eutrophication, and 0.18 kg PM_{2.5}-eq/t FW of particulate matter up to 2.5 μm diameter. Moreover, emissions from post-composting, energy consumptions of drying and chemical fertilizer substitution ratio were detected by contribution analysis as the main contributors to those impacts. Finally, sensitivity analysis indicated that the substitution ratio of mineral fertilizer and protein feed as well as energy consumption were the most influential parameters, therefore control of the post-composting process of residual material should be closely monitored because it was responsible for significant environmental load caused by N-related emissions.

© 2020 Elsevier B.V. All rights reserved.

1. Introduction

Food waste (FW) is a global issue; 1.3 billion tons of FW is generated per year, which amounts to one-third of the food produced for human consumption (FAO, 2018). There is also a growing demand for protein feed because of the increasing world population (Huis et al., 2013). To

* Corresponding author at: School of Environment, Tsinghua University, Beijing 100084, China.

E-mail address: luwenjing@tsinghua.edu.cn (W. Lu).

stabilize the feed price and lower the environmental pressures, edible insects used for human food and feed are recommended as an alternative solution by the Food and Agriculture Organization (FAO, 2013).

Black soldier fly (BSF, *Hermetia illucens*), a saprophytic insect, is widely distributed in tropical, subtropical and warm temperate regions of the world (Kumar et al., 2018). There are four stages in the BSF growth cycle: egg, larva, pupa and adult. The larvae can reach up to 220 mg in weight when fed with organic materials (Salomone et al., 2017). Studies have shown that BSF larvae (BSFL) can convert FW (Win et al., 2018), human feces (Lalander et al., 2018), and animal manure (Li et al., 2011) into their own body mass. BSFL bioconversion could be affected by different substrates. The rearing substrate has effects on growth performance, waste reduction efficiency and chemical composition of BSFL (Lim et al., 2019; Meneguz et al., 2018). Poultry feed, dog food, FW, fruit and vegetables, abattoir waste, poultry manure and other wastes could be the substrates for BSFL bioconversion; the pre-pupae grown on these substrates were quite large in all cases (>210 mg/larva) except for those reared on poultry manure (165 mg/larva) (Lalander et al., 2019).

The BSF pre-pupae are rich in nutrients and can partly replace the animal feed for fish and poultry (Barragan-Fonseca et al., 2017). Because of their high oil content, BSF pre-pupae could also be considered a potential material for biodiesel production (Wong et al., 2019; Zheng et al., 2012). When BSFL bioconversion was compared with aerobic composting, it was found to reduce the treatment time by up to 1.6-fold (Cai et al., 2018). Importantly, BSF adults are not attracted to human habitats or foods, neither are they a vector in disease transmission (Lalander et al., 2019). Therefore, bioconversion of organic waste by BSFL is potentially an attractive alternative for waste valorization in the new century and has recently attracted great attention.

The focus of previous studies was the potential use of BSFL for biodiesel and animal feed (Surendra et al., 2016; Win et al., 2018), the digestion of different types of waste (Beskin et al., 2018) as well as the efficacy of BSFL bioconversion (Jiang, 2018; Li et al., 2018). However, limited research has been conducted on the environmental impact from a life cycle perspective. Smetana et al. (2016) conducted a life cycle assessment (LCA) study of insect production at the industrial level, which indicated that 2–5 times greater environmental benefits could be achieved by using insect-based protein powder and meat substitution when compared with traditional products. However, only the information on materials (e.g. wheat bran, water, barley grain, minerals) and energy consumption (e.g. electrical, heat) were provided, while the gaseous emissions were not tested.

Salomone et al. (2017) and Allegrretti et al. (2018) conducted LCA studies on the bioconversion of BSFL from pilot plants located in Southern Italy and Brazil, respectively. However, the emission data were adopted from house fly larvae (van Zanten et al., 2015). Mertenat et al. (2019) presented a study that assessed the global warming potential (GWP) of a BSFL treatment facility in Indonesia. Direct methane (CH₄) and nitrous oxide (N₂O) emissions from this facility were measured. The BSFL facility was compared with an open windrow composting facility; however, because of the non-transparent database (composition of food waste, energy structure, transformation of substances, etc.), there are still some limitations when comparing with other technologies.

The life cycle impact of a systemized pilot plant of BSF insects was assessed with multi-season datasets (Smetana et al., 2019); the results demonstrated that both fertilizer and insect production were favorable in terms of environment impact when compared with many conventional organic fertilizers and animal- or plant-based proteins.

Performance analysis of insect biorefinery of three food waste sources (brewery grains, potato peels, expired food) from the life cycle perspective was conducted (Ites et al., 2020); the results indicated that system performance from both the LCA and economics standpoint were highly dependent on the type of waste.

Material flow analysis (MFA) is important and necessary for LCA and often forms the basis of life cycle inventory (LCI) (Allesch and Brunner,

2017; Jensen et al., 2017). MFA can give not only an insight into the material and substance transfer and distribution in a defined area, but also considers the material and substance flow from a systematic perspective. It is widely applied in many fields and disciplines (Klinglmair et al., 2016; Laner et al., 2016; Steindl-Rast and Lebell, 1997; Zhang et al., 2016). MFA can be utilized when designing a waste management system (Allesch and Brunner, 2017).

MFA is an important basis for system analysis, such as LCA. However, studies seldom present and interpret the material flow in FW treatment systems in detail. Guo et al. (2019) conducted MFA of alternative biorefinery scenarios for FW management, but the technologies related to insects were not included.

In the present study, we collected the gaseous emissions and solid samples from a FW-based BSFL bioconversion plant and linked these values to a systematic mass balance. Our study aimed to: (1) assess the material flow inside the BSFL bioconversion system from a systematic perspective; (2) provide a full LCI (including the detailed substance transfer coefficients) of the BSFL bioconversion system; and (3) reveal the environmental impact of a BSFL bioconversion plant.

2. Materials and methods

2.1. FW-based BSFL bioconversion plant

The study was conducted in a FW-based BSFL bioconversion plant located in Hangzhou, China, with the capacity to treat 15 tons of FW (wet weight, WW) per day. A schematic diagram of the plant is shown in Fig. 1. The plant received FW from households, restaurants, and canteens. The moisture content of raw FW was 76%. FW was crushed and mixed with rice hull powder at a ratio of 4:1 (WW), which was hereafter referred to as food waste feed (FWF). The moisture content of FWF was 62% and the carbon (C) to nitrogen (N) ratio (C/N) was 19. Rice hull powder acted as the bulking agent to improve air permeability and thus optimize the composting conditions during BSFL growth.

After two days at room temperature (around 25 °C), the mixture became pasty, which was suitable for BSFL bioconversion. The composition of materials in this study was determined and presented in Table 1. The moisture was measured on a WW basis, while other components (i.e., volatile solids, C, N, protein and lipid) were based on total solids (TS).

In this plant, a total of 600 boxes each with a volume of 30 × 26 × 17 cm³ were used for BSFL bioconversion. Three of these boxes were chosen as parallels ($n = 3$) for controlled MFA experiment. Approximately 14,000 4-day-old larvae were placed in each box.

The 14,000 4-day-old larvae were counted using the following steps:

- (1) First, we picked out 10–20 larvae, weighed them on an electronic analytical balance, and calculated the average mass of a single larva:

$$m_{\text{one-larva}} = \frac{m_{n-\text{larva}}}{n} \quad (1)$$

where $m_{\text{one-larva}}$ is the average mass of a single larva; $m_{n-\text{larva}}$ is the total weight of n larvae, and n is the number of larvae weighed.

- (2) The first step was repeated 4–5 times to improve the estimate of the average mass of a single larva:

$$m_{\text{one-larva-average}} = \frac{m_{\text{one-larva-1}} + m_{\text{one-larva-2}} + \dots + m_{\text{one-larva-p}}}{p} \quad (2)$$

where $m_{\text{one-larva-average}}$ is the average mass of one larva; and p is the number of repetitions of step (1);

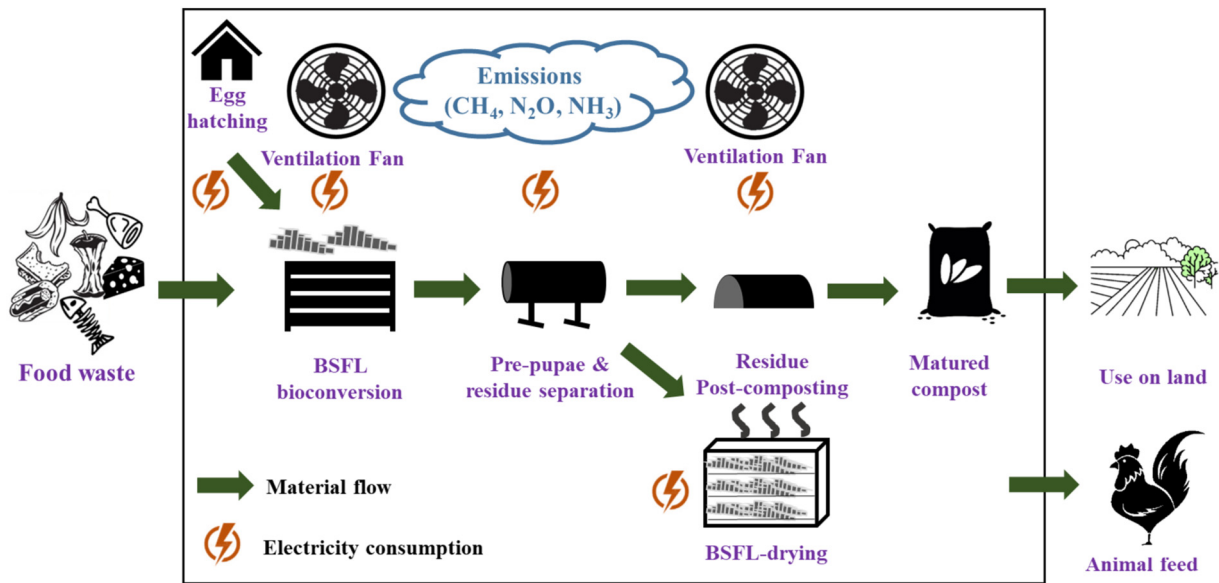


Fig. 1. A scheme diagram of food waste-based BSF bioconversion plant.

- (3) The average mass of a single larva was used to calculate the total mass of the desired number of larvae:

$$m_{Q-larvae} = m_{one-larva-average} \times Q \quad (3)$$

where $m_{Q-larvae}$ is the total mass of the desired number of larvae; and Q is the desired number of larvae.

FWF, composed of FW and rice hull powder at a ratio of 4:1, was added every day. The total amount was 25 kg per each box, and the feeding rate was shown in Fig. 2 as described in a previous study (Jiang et al., 2019).

The initial weight of each larva was 0.85 ± 0.06 mg. BSFL bioconversion was considered complete when one quarter of the larvae turned to pre-pupae, which usually took 10 to 15 days. BSF pre-pupae (the final larval stage) and residue were then separated. A vibration separator machine was used for pre-pupae and residue separation according to the varied particle sizes. There was a screening deck in the separator; when the machine was in operation, only pre-pupae were left on the screening deck. The separated residue was considered to be immature compost and needed to be matured through post-composting before land application, indicated as post-composting (Gold et al., 2018). The final products were the pre-pupae and matured compost. BSF pre-pupae were directly dried in an oven at 105°C for at least 12 h. The energy sources adopted from the international database were consistent with the local energy structure (74% of hard coal, 7% of hydro and 19% of natural gas).

2.2. System boundary and sample collection

The functional unit was defined as bioconversion of 1 ton of FW. There were three paths of material flow in the process of FW

bioconversion by BSFL: emissions to the environment, bioconversion to BSF pre-pupae, and matured compost. Environmental benefits and burdens from the upstream and downstream processes related to the bioconversion as well as the products substitution ratio were also taken into consideration. However, the processes of waste collection, transport, products distribution, and the construction of the plant were excluded.

To investigate the MFA of BSFL bioconversion of FW, a sampling schedule of both solid and gaseous samples was carried out. Operational data were acquired from the plant, including the overall mass balance and the consumption of electricity and additional materials.

2.2.1. Solid samples

In the 10-day bioconversion procedure, BSFL were fed FWF as shown in Fig. 2 every day. Substrates were mixed homogeneously before daily sampling, and one sample (0.2 kg) was taken from each box. All the solid samples were stored at -20°C .

Moisture, volatile solids (VS), total C, and total N were measured according to standard procedures (Bareither et al., 2010). The concentrations of lipid and protein were determined by using a Soxhlet device and the Kjeldahl method, respectively (Li et al., 2017; Naumann et al., 1997; Song et al., 2019).

2.2.2. Gaseous samples

For gaseous sampling, the BSFL bioconversion plant was closed (operating under negative pressure) and the gases produced from this BSFL bioconversion plant were discharged from 12 fan-equipped outlets installed on the plant walls. The specialized sampler (Dalian Delin Gas Packing Co., Ltd., ASP-3) consisted of a sampling bag, sampling tube and a vacuum pump as described elsewhere (Zhao et al., 2011). We collected gaseous samples for 3 days in

Table 1
Composition of food waste and rice hull powder.

	Water % wet weight	Volatile solid % total solid	Carbon % total solid	Nitrogen % total solid	Protein % total solid	Lipid % total solid
Food waste	76.09 ± 0.74	89.01 ± 1.33	57.52 ± 0.27	4.10 ± 0.03	25.88 ± 0.02	44.17 ± 0.10
Rice hull powder	6.04 ± 0.02	80 ± 0.85	40.40 ± 0.03	1.00 ± 0.01	6.25 ± 0.05	3.74 ± 0.04
Food waste feed (the mixture)	62.08	84.54	49.04	2.57	16.15	24.13

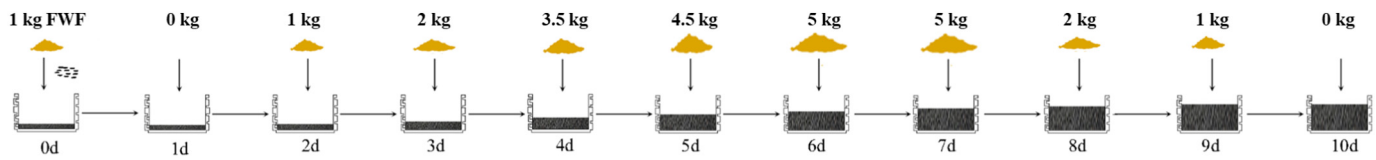


Fig. 2. Daily feeding procedure for BSFL bioconversion (Jiang et al., 2019).

this FW-based BSFL bioconversion plant. For each sampling day gaseous samples were taken every 4 h, i.e., 4:00, 8:00, 12:00, 16:00, 20:00, 24:00, and in total 18 samples were obtained over a period of three days. Samples were transported to the lab and analyzed within 48 h. CH₄ and N₂O were analyzed by gas chromatography (Techncomp, 7890 II) with the limit of detection of 100 ppm for CH₄ and 0.05 ppm for N₂O; NH₃ was measured by an ammonia detector (LH-2000-NH₃, Qingdao Liheng Environmental Protection Equipment Co., Ltd), with the accuracy of 0.01 ppm and the upper limit of detection of 100 ppm. The mass of gases emitted (i.e., CH₄, N₂O and NH₃) was calculated using Eq. (4).

$$m = \frac{c \times S \times v \times 3600 \times H \times M}{1000 \times V_m} \quad (4)$$

where m is the mass of the target gas (CH₄, N₂O and NH₃) with environmental impacts, g/d; v is the wind speed, m/s; S is the ventilation area, m²; H is the ventilation time, h; c is the gas concentration, ppm; M is molecular weight of gas, g/mol; and V_m is the standard molar volume of gas, 22.4 L/mol.

2.3. Mass balance and LCI

The BSFL bioconversion process was assessed using the waste-LCA model EASETECH (Environmental Assessment System for Environmental Technologies, v2.8.5), which is one of the most widely used LCA models around the world, especially for solid waste because it is good at dealing with heterogeneous materials. The development and function of this model is detailed in Claveul et al. (2014). More detail can be found in previous studies (Andersen et al., 2012; Bernstad and la Cour Jansen, 2011; Hansen et al., 2006; Starostina et al., 2018; Zhao et al., 2015). Mass and substance flow analysis was performed using the mass balance model STAN (substance flow ANALysis, 2.6) (Cencic and Rechberger, 2008). The MFA was the basis for the LCI.

The LCI involved input waste, additive materials, emissions, and energy consumption of each subprocess as well as the composition of final products; it was based on the principle of mass balance. The LCA was assessed using the results of the LCI in EASETECH.

The LCI of BSFL bioconversion is shown in Tables 2 and 3. All the inventory data were collected and measured from this plant excluding the emissions of CH₄ and N₂O. The values for CH₄ and N₂O emissions during BSFL bioconversion were taken from the literature (Mertenat et al., 2019) (Table 2). We assumed that the post-composting of residue showed similar emissions to general organic waste from household or kitchen waste (Table 3). This assumption was also used in Mertenat et al. (2019).

We assumed that products could be utilized under the following conditions: the insect protein contained in BSF pre-pupae could provide 100% of the protein feed on a protein basis, while matured compost could provide 60% of the N in chemical fertilizer (empirical data from the plant manager).

2.4. Life cycle impact assessment (LCIA)

The results of LCI can be used to assess the life cycle impact (e.g., on the environmental impacts). Five impact categories: GWP, acidification, terrestrial eutrophication, marine eutrophication, and particulate

matter were chosen according to the related inventory. Characterization factors of the related inventories in the chosen impact categories were obtained from the International Reference Life Cycle Data System (ILCD) Handbook (Laurent et al., 2014).

Table 2

Life cycle inventory data for BSF bioconversion of food waste. All data were derived from the present study unless specified otherwise.

	LCI data	Amount	Unit
Input waste	Food waste	1000	kg
Materials consumption	Rice hull powder	250	kg
Energy consumption	Egg hatching	13.287	kWh/t-BSF egg
	Crushing	14.71	kWh/t-FWF
	Ventilation	2.35	kWh/t-FWF
	Bioconversion	2.59	kWh/t-FWF
	Pre-pupae & residue separation	31.48	kWh/t-WW mass for separation
	Post-composting	0.89	kWh/t-WW mass for post-composting
	BSF pre-pupae drying	281.90	kWh/t-TWW mass for drying
Emissions during bioconversion	C to atmosphere	62.70	%, C _{initial} of FW
	C to BSF pupae	10.35	%, C _{initial} of FW
	N to atmosphere	50.86	%, N _{initial} of FW
	N to BSF pupae	7.71	%, N _{initial} of FW
	CH ₄ -C _{emitted}	3.50E-04 ^a	%, C _{emitted} of FW
		4.04E-04	kg/t-FW
	NH ₃ -N _{emitted}	2.19	%, N _{emitted} during bioconversion
		0.13	kg/t-FW
	N ₂ O-N _{emitted}	0.11 ^a	%, N _{emitted} during bioconversion
		8.64E-03	kg/t-FW
Emissions during post-post-composting	C to atmosphere	33.66	%, C _{initial} for post-composting
	N to atmosphere	34.31	%, N _{initial} for post-composting
	CH ₄ -C _{emitted}	1.40 ^b	%, C _{emitted} during post-composting
		0.26	kg/t-FW
	NH ₃ -N _{emitted}	89.24	%, N _{emitted} during post-composting
		1.59	kg/t-FW
	N ₂ O-N _{emitted}	1.24 ^b	%, N _{emitted} during post-composting
		2.85E-02	kg/t-FW
Final products	Dried BSF pre-pupae	30.94	kg
	Matured compost	300	kg
Composition of fresh BSF pre-pupae	Water	67	%, TWW
	VS	82	%, TS
	Protein	41	%, TS
	Lipid	47.30	%, TS
Composition of matured compost	Water	20	%, TWW
	N	1.26	%, TS
	P ₂ O ₅	1.08	%, TS
	N-NH ₄ ⁺	1.40E-02	%, TS
	K ₂ O	0.46	%, TS
	As	0.14	mg/kg
	Hg	5.80E-03	mg/kg
	Cr	24.20	mg/kg

^a The ratios of emitted carbon transferred into CH₄ and emitted nitrogen transferred into N₂O during BSF bioconversion were collected from (Mertenat et al., 2019), while the results from present study were less than 3.46% and 0.25% respectively.

^b The gases emissions (CH₄ and N₂O) from post-composting process were collected from Table 3.

Table 3Data reference of CH₄ and N₂O emission from organic waste composting.

Materials	Emitted as CH ₄	Emitted as N ₂ O	Refs
	%, C _{emitted}	%, N _{emitted}	
Food waste	1.70	0.40	Hodge et al., 2016
Organic household waste	–	0.77	Chung, 2007
Organic household waste	1.00	2.00	(Bernstad and la Cour Jansen, 2011)
Kitchen waste	1.50	1.80	Boldrin et al., 2010
Average ± SD	1.40 ± 0.29	1.24 ± 0.67	

2.5. Sensitivity analysis

Sensitivity analysis was conducted to assess how sensitive the life cycle impacts were to changes in specified parameters, using the sensitivity ratio (SR) (Zhao et al., 2011). The approach as shown in Eq. (5) was described in detail elsewhere (Clavreul et al., 2012). All parameters were decreased by 10% for sensitivity analysis. A parameter appeared as a sensitive parameter when the absolute value of SR was higher than 1.

$$SR = \frac{\frac{result_{changed} - result_{initial}}{parameter_{changed} - parameter_{initial}}}{\frac{result_{initial}}{parameter_{initial}}} \quad (5)$$

where $parameter_{initial}$ is the initial value related to one parameter; $parameter_{changed}$ is the changed value of that parameter; $result_{initial}$ is the impact result with $parameter_{initial}$, and $result_{changed}$ is the impact result with $parameter_{changed}$.

3. Results and discussion

3.1. MFA

One ton of FW coupled with 0.25 ton of rice hull powder yielded 31 kg of dried BSF pre-pupae and 300 kg of matured compost (WW) after BSFL bioconversion and post-composting, with the remainder emitted as exhaust gas to the environment. These outputs were consistent with Salomone et al. (2017), where 30 kg of dried larvae and 335 kg of matured compost were obtained from a similar BSF processing of 1 t of FW.

The initial water, C and N contents were 64% WW, 48% TS, and 2.35% TS, respectively, whereas those in the BSFL residue were 35% WW, 41% TS and 2.04% TS, respectively, after 10-day bioconversion (Fig. 3). When treating FW by BSFL in Liu et al. (2020), N content increased sharply and was consistent with the results of our study, increasing from 2.69% TS to 3.19% TS, with 3.02% TS in the BSFL residue.

Fig. 4 presents the material flows during BSFL bioconversion in this study. We assumed that rice hull powder acted only as a bulking agent in FWF and was not substantially taken up by BSFL. Focusing on the outputs from the bioconversion process, the total input of FWF

(FW + rice hull powder) (WW) was distributed into BSF pre-pupae (10%), BSF residue (40%) and gaseous emissions (50%). Because of the properties of the separator, three-quarters of pre-pupae were separated and one-quarter remained in the residue. The BSF pre-pupae drying and the residue post-composting were done after separation. Therefore, the material distribution (in WW) of the whole process (including BSFL bioconversion, pre-pupae and residue separation and residue post-composting) was 8% to fresh pre-pupae, 24% to the matured compost and 68% to the environment as gaseous emissions (50% from the bioconversion process plus 18% from post-composting). The distribution based on dry matter was 43% to emissions, 6% to BSF pre-pupae and 51% to the matured compost.

The material flow of total mass, water, C and N in the BSFL bioconversion system is shown in Fig. 4. A significant proportion of water (84% of total water) was emitted to the environment, 8% was transformed to fresh BSF pre-pupae, and 8% remained in the mature compost. C flow analysis showed that 43% of C in FWF was emitted to the environment, 52% remained in matured compost, and 5% was stored in pre-pupae. N flow analysis showed that 53% of N was emitted to the environment, 42% was kept in matured compost, and 5% was stored in pre-pupae, which is in accordance with the findings by Lalander et al. (2014). During BSFL conversion of pig manure of different moisture content, the N in BSFL of all the investigated treatments accounted for 1.03–12.67% of the initial total N (Chen et al., 2019).

The results of our study demonstrated that bioconversion of FWF by BSFL could achieve a significant reduction in waste mass as well as individual components, which is in accordance with the findings in other studies. It was reported that the reduction in fresh mass was 68% for municipal organic waste (Diener et al., 2011). Rehman et al., 2017 also showed reductions of water, C and N of 73%, 52.4–71.9% and 24–56% respectively, in organic waste treated by BSFL. An average reduction of 49% in dry mass of polluted corn leaves was achieved by BSFL (Bulak et al., 2018). Reductions of 55% N and 62% C were achieved by Newton et al. (2005).

3.2. LCI

LCI of the BSFL bioconversion (Table 2) was conducted in the current study, including material consumption, energy consumption, transfer coefficients of each process, the main gaseous emissions, as well as the

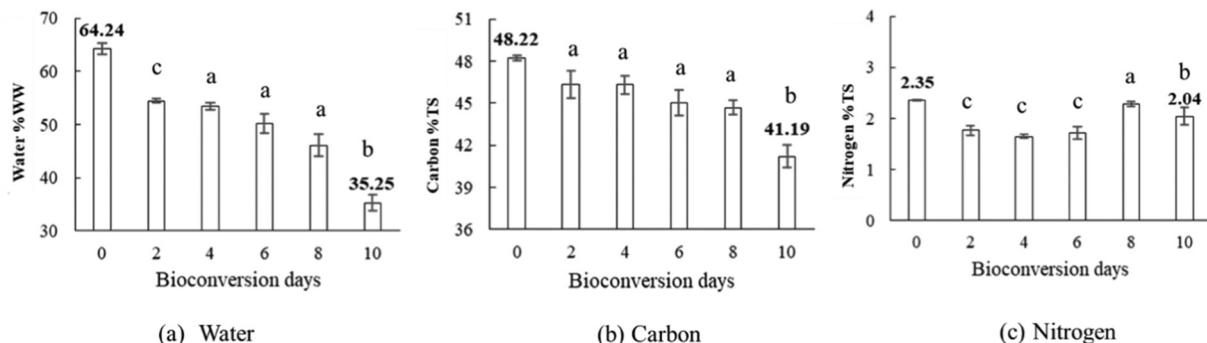
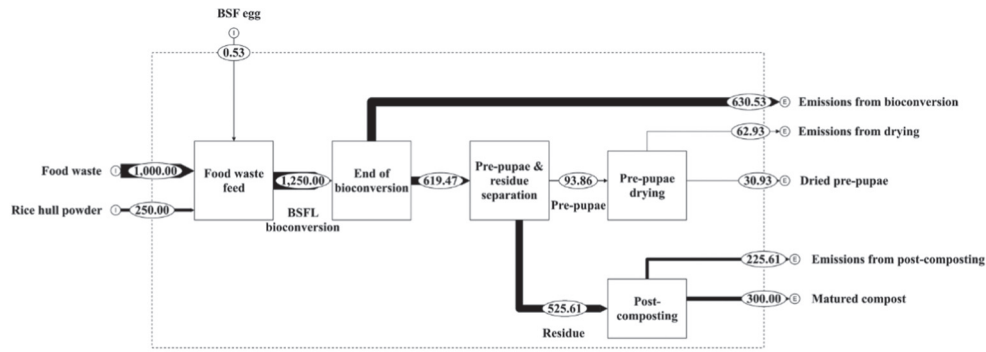
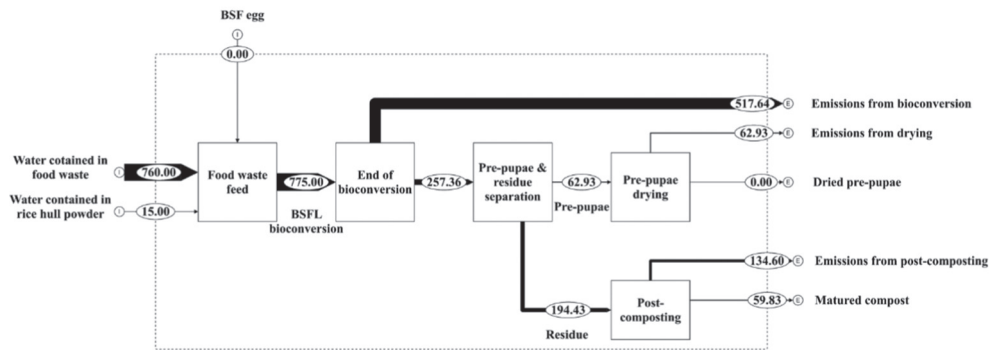


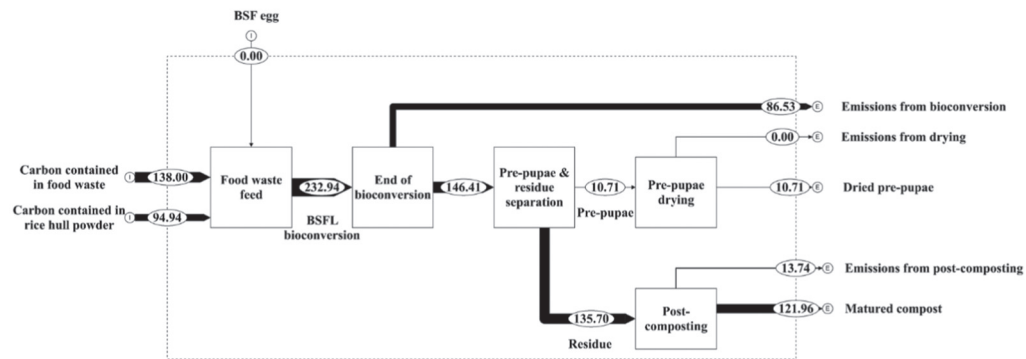
Fig. 3. Variation of BSFL substrate in the course of bioconversion (Equal letter represent no significant difference from one day to the other).



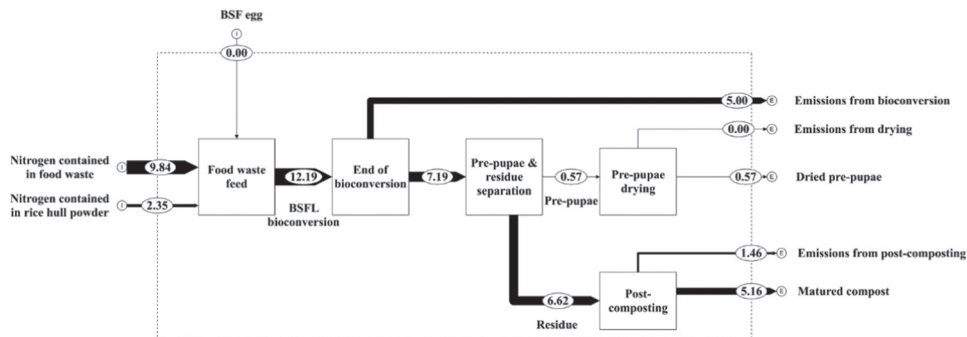
(a) Total mass flow analysis



(b) Water flow analysis (Water: FWS, 62.00% wet weight)



(c) Carbon flow analysis (Carbon: FWS, 18.64% wet weight)



(d) Nitrogen flow analysis (Nitrogen: FWS, 0.98% wet weight)

Fig. 4. Material flows of the food waste-based BSFL bioconversion system.

composition of the final products. NH_3 emissions from both BSFL bioconversion and residue post-composting were measured in this study. The concentrations of CH_4 and N_2O were lower than the detection limits (100 ppm for CH_4 and 0.05 ppm for N_2O) in the present study. As mentioned above, the values for CH_4 and N_2O emissions during BSFL bioconversion were taken from the literature (Mertenat et al., 2019) (Table 2). We assumed that the post-composting of residue showed similar emissions to general organic household waste (Table 3), as used in Mertenat et al. (2019).

When 1 ton of FW was treated by BSFL, 0.26 kg of CH_4 , 1.72 kg of NH_3 and 0.037 kg of N_2O could be emitted during the whole process (includes both BSFL bioconversion and post-composting). The post-composting process after BSF pre-pupae bioconversion was the main source of gaseous emission, which was consistent with Mertenat et al. (2019).

LCI showed that 51% of N contained in FW was emitted to the environment in the bioconversion process, and 2% of the emitted N was in the form of NH_3 . Thus, 1.02% of the initial N contained in FW was emitted as NH_3 during the bioconversion procedure. When converting pig manure by BSFL, N loss in the form of NH_3 accounted for about 1.47–2.36% of the initial total N and N_2O emission was minimal (Chen et al., 2019). While NH_3 emissions were very low in the present study, no NH_3 emissions were detected by Ermolaev et al. (2019) who performed fly larvae composting in 11 cm deep small boxes with 700 larvae.

During the post-composting process, 34% of N contained in the solids used for post-composting was emitted, mainly as NH_3 . Our study demonstrated that the emissions of NH_3 from the BSFL bioconversion and the post-composting processes were 0.13 kg/t and 1.59 kg/t FW, respectively, which was consistent with the results of a previous BSFL study demonstrating NH_3 emission of 0.15–1.69 kg/t from BSFL bioconversion (Pang et al., 2020).

Energy consumption was measured for the whole process, including egg hatching, FW crushing, BSFL bioconversion, products and residue separation, BSF pre-pupae drying as well as the post-composting process (Table 2). The highest energy consumption occurred in the BSF pre-pupae drying process, accounting for 35% of the total energy consumption.

The composition of the final products of bioconversion, i.e., BSF pre-pupae and matured compost, is shown in Table 2. BSF pre-pupae and matured compost can be used as substitutes for animal feed and chemical fertilizer, respectively. The BSF pre-pupae were rich in protein and lipid ($41 \pm 0.43\%$ TS and $47 \pm 1.51\%$ TS, respectively). This would be acceptable as fish meal or poultry nutrition (Henry et al., 2015; Khan, 2018; Liu et al., 2017). When used as feed, the mass ratio of BSF dried pre-pupae was recommended as 10% (WW) of the weight of feed (Huailong and Xiaoyan, 2018; Shisheng, 2016). We assumed that the insect protein contained in BSF pre-pupae could substitute for 100% of the protein feed on a protein basis. The upstream process of producing protein feed was selected from Ecoinvent (Ecoinvent, 2007) with the protein content of 100% TS. The heavy metals content as well as quality were within the standards for organic fertilizer in China (NY 525-2012). Matured compost resulting from the bioconversion described here was estimated to substitute 60% of chemical fertilizer on a N basis (empirical data from the plant manager).

3.3. LCIA

The overall GWP of the BSFL bioconversion system was 17.36 kg $\text{CO}_2\text{-eq/t}$ FW (Fig. 5 (a)). The specific quantities were -72.69 kg $\text{CO}_2\text{-eq/t}$ FW for the utilization of products, 17.24 kg $\text{CO}_2\text{-eq/t}$ FW from emissions and 72.81 kg $\text{CO}_2\text{-eq/t}$ FW for energy consumption. Energy consumption and greenhouse gas emissions during the whole process were taken as loads to the environment. Hence, when they were balanced with the substitution of raw materials or products from the BSFL bioconversion, CO_2 net emissions were saved. Results from

Salomone et al. (2017) showed emission from each ton of FW of 30.2 kg CO_2 eq in terms of GWP. The reduction in environmental impacts when compared with conventional composting was mostly caused by replacement of the production of N fertilizers.

Drying accounted for the largest proportion (35%) of the total energy consumption. Van Zanten et al. (2015) and Mertenat et al. (2019) also demonstrated that energy consumption was the main contributor to GWP impact in the system of either BSFL or housefly larvae growing on FW. Salomone et al. (2017) also showed that high GWP impacts were caused by electricity consumption during the production of feed and compost, whereas drying was associated with the most significant energy consumption in the bioconversion system. The GWP caused by emissions from the post-composting process was 5-fold higher than that from BSFL bioconversion. N_2O emitted during the post-composting process was the main contributor to the a GWP of 310 kg $\text{CO}_2\text{-eq}$ (Hayes et al., 2016).

Acidification was commonly related to the pollution caused by N-compounds. The comprehensive acidification impact was 5.54 kg $\text{SO}_2\text{-eq/t}$ FW. The emissions from post-composting contributed most to the acidification impact (Fig. 5 (b)). The high NH_3 emission during the post-composting process led to a serious acidification load. The acidification related to products utilization was -0.11 kg $\text{SO}_2\text{-eq/t}$ FW, and this saving on acidification impact was mainly because of the avoidance of production of the equivalent amount of inorganic fertilizer. Nevertheless, when compared with the post-composting process, the impacts related to products substitution and energy consumption were insignificant.

The total impact on terrestrial eutrophication related to NH_3 emission was 24.05 mol N-eq (Fig. 5 (c)). The largest burden came from the emissions during the post-composting process, with 21.36 mol N-eq mainly caused by the emission of NH_3 .

The impact on marine eutrophication was associated with NH_4^+ generation and release to the environment (Peigné and Girardin, 2004). The total impact on marine eutrophication was 0.54 kg N-eq (Fig. 5 (d)). The marine eutrophication caused by products utilization was 0.32 kg N-eq, 0.16 kg N-eq in emissions and 0.064 kg N-eq in energy consumption. Analysis of the contributing factors showed that the largest burden came from the utilization of compost because of the NH_4^+ conversion to nitrate that was removed via leaching and run-off.

The overall particulate matter for the system was 0.18 kg $\text{PM}_{2.5}\text{-eq/t}$ FW (Fig. 5 (e)). Particulate matter was associated with emissions and energy consumptions of 0.11 and 0.08 kg $\text{PM}_{2.5}\text{-eq/t}$ FW, respectively. The substitution and utilization of feed and matured compost resulted in -0.02 kg $\text{PM}_{2.5}\text{-eq/t}$ FW savings. NH_3 emissions from post-composting was the main contributor to particulate matter. When compared with the loadings from emissions and energy consumption, the saving by product utilization was negligible.

The results on the impacts of acidification, terrestrial eutrophication, marine eutrophication, and particulate matter in this study filled an important knowledge gap in the environmental impact of the BSFL bioconversion system. Contribution analysis could help us find the “hot spots” in the chosen environmental categories. The top three processes with respect to the GWP were chemical fertilizer substitution, energy consumption of drying and emissions from post-composting. For acidification and terrestrial eutrophication, the most important component was the emissions from post-composting. In marine eutrophication, chemical fertilizer substitution played an important role followed by emissions from post-composting and protein feed substitution. Emissions from post-composting, energy consumption for drying and bioconversion were the main contributors for particulate matter.

Post-composting and chemical fertilizer substitution were recognized as the key processes in contribution analysis, which is consistent with the conclusions of other's study (Salomone et al., 2017). Therefore, greater attention should be paid to the emissions from post-composting and the utilization of products. The results from the hot spots could motivate us to find better ways to deal with the BSFL residues from bioconversion.

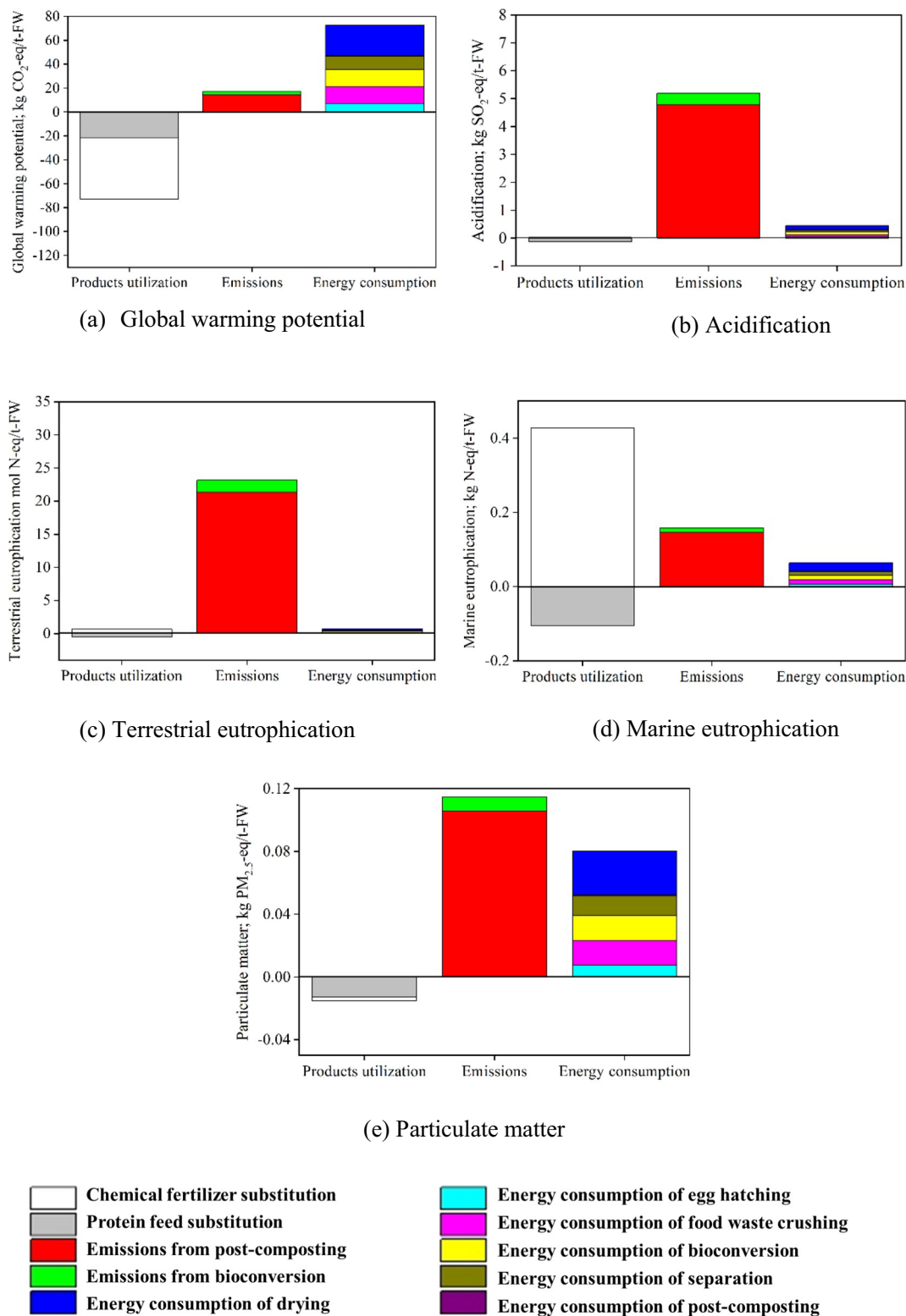


Fig. 5. Life cycle characterized system contributions to the global warming potential, acidification, terrestrial eutrophication, marine eutrophication and particulate matter.

3.4. Sensitivity analysis

The sensitivity ratio of 34 parameters from the five chosen environmental categories was calculated to reveal the influential parameters in the FW-based BSFL bioconversion system (Table 4).

Generally, products utilization and substitution as well as energy consumption were the most sensitive processes in terms of GWP; while the process of NH₃ emission to ambient air both during bioconversion and post-composting processes had greater impacts on acidification and terrestrial eutrophication. The GWP impact was highly

Table 4

Sensitivity ratios of significant parameters with the absolute value higher than 0.9 in the five chosen impact categories global warming potential (GWP), acidification (AC), terrestrial eutrophication (Eut), marine eutrophication (Eum) and particulate matter (PM).

	GWP	AC	Eut	Eum	PM
Nitrogen emitted to ambient during bioconversion	–	–1.0	–1.0	–	–
Substitution of mineral fertilizer	–2.8	–	–	–	–
Energy consumption for drying	1.5	–	–	–	–
Substitution of protein feed	–1.3	–	–	–	–
Energy consumption for separation	1.1	–	–	–	–
Nitrogen emitted as NH ₃ during post-composting	–	0.9	0.9	–	–
Nitrogen emitted during post-composting	–	0.9	0.9	–	–

influenced by the substitution proportion of chemical fertilizer and protein feed, mainly because the avoidance of production and usage of chemical fertilizer. N emitted to the environment during bioconversion was a sensitivity parameter for acidification and terrestrial eutrophication. Acidification was influenced by the N-related emissions during post-composting.

The sensitivity analysis indicated that high N content in the material used for post-composting was an important issue, resulting in the environmental loads of acidification and eutrophication. This could be balanced through technical improvements, such as improving the management of BSFL bioconversion to achieve higher accumulation of N in BSFL pre-pupae, increasing the pre-pupae recovery rate during separation, decreasing NH₃ emissions by adding diatomite (Ren et al., 2019) or biochar amendment (Wu et al., 2019), and applying an acid scrubber or compost biofilter (Wang et al., 2019) to mitigate NH₃ emissions during post-composting. Utilizing fresh instead of dried BSFL pre-pupae to reduce the energy consumption is an alternative way to decrease the impacts because changes in energy consumption greatly impact the GWP.

4. Conclusions

This study explored the environmental impacts (GWP, acidification, terrestrial and marine eutrophication, and particulate matter) of FW bioconversion by BSFL using MFA. Totals of 31 kg of dried BSFL pre-pupae (from 92 kg of WW pre-pupae) and 300 kg of matured compost (WW) were obtained from the bioconversion of 1 ton of FW coupled with 0.25 ton of rice hull powder (WW, with 62% moisture content in the mixture) by BSFL. In terms of elements, 48% of C and 58% of N were transformed into BSFL and emitted to the environment. The process of pre-pupae drying caused the largest energy consumption, which was strongly related to the environmental impact measured by GWP. LCA showed that the GWP of BSFL bioconversion was 17.36 kg CO₂-eq/t FW, 5.54 kg SO₂-eq/t FW for acidification, 24.05 mol N-eq/t FW for terrestrial eutrophication, 0.54 kg N-eq NH₃/t FW for marine eutrophication, and 0.18 kg PM_{2.5}-eq/t FW for particulate matter. A detailed LCI, covering the comprehensive transfer coefficients of BSFL bioconversion, provided in this study could be used for comparisons with other similar FW disposal plants.

CRedit authorship contribution statement

Hanwen Guo: Conceptualization, Methodology, Software, Investigation, Data Curation, Writing-Original Draft

Chengliang Jiang: Investigation, Validation

Zhijian Zhang: Conceptualization, Resources, Writing-Review & Editing

Wenjing Lu: Supervisor, Conceptualization, Writing-Review & Editing, Funding acquisition

Hongtao Wang: Supervisor

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This present study was supported by the National Key Research and Development Program of China (Grant No. 2018YFD1100600). The authors would also like to express their sincere gratitude to the manager and workers of Hangzhou Gusheng Biotechnology Co., Ltd., and to Dr. Yuyang Long, School of Environmental Science and Engineering, Zhejiang Gongshang University, for their helpful assistance during the sample collection campaign.

References

- Allegretti, G., Talamini, E., Schmidt, V., Bogorni, P.C., Ortega, E., 2018. Insect as feed: an emergency assessment of insect meal as a sustainable protein source for the Brazilian poultry industry. *J. Clean. Prod.* 171, 403–412. <https://doi.org/10.1016/j.jclepro.2017.09.244>.
- Allesch, A., Brunner, P.H., 2017. Material flow analysis as a tool to improve waste management systems: the case of Austria. *Environ. Sci. Technol.* 51, 540–551. <https://doi.org/10.1021/acs.est.6b04204>.
- Andersen, J.K., Boldrin, A., Christensen, T.H., Scheut, C., 2012. Home composting as an alternative treatment option for organic household waste in Denmark: an environmental assessment using life cycle assessment-modelling. *Waste Manag.* 32, 31–40. <https://doi.org/10.1016/j.wasman.2011.09.014>.
- Bareither, C.A., Breitmeyer, R.J., Meyer, L.L., Benson, C.H., Edil, T.B., Barlaz, M.A., 2010. Physical, chemical, and biological characterization of solid waste samples. *2nd Glob. Waste Manag. Symp* 1–9.
- Barragan-Fonseca, K.B., Dicke, M., van Loon, J.J.A., 2017. Nutritional value of the black soldier fly (*Hermetia illucens* L.) and its suitability as animal feed - a review. *J. Insects Food Feed* 3, 105–120. <https://doi.org/10.3920/JIFF2016.0055>.
- Bernstad, A., la Cour Jansen, J., 2011. A life cycle approach to the management of household food waste - a Swedish full-scale case study. *Waste Manag.* 31, 1879–1896. <https://doi.org/10.1016/j.wasman.2011.02.026>.
- Beskin, K.V., Holcomb, C.D., Cammack, J.A., Crippen, T.L., Knap, A.H., Sweet, S.T., Tomberlin, J.K., 2018. Larval digestion of different manure types by the black soldier fly (Diptera: Stratiomyidae) impacts associated volatile emissions. *Waste Manag.* 74, 213–220. <https://doi.org/10.1016/j.wasman.2018.01.019>.
- Boldrin, A., Hartling, K.R., Laugen, M., Christensen, T.H., 2010. Environmental inventory modelling of the use of compost and peat in growth media preparation. *Resour. Conserv. Recycl.* 54, 1250–1260. <https://doi.org/10.1016/j.resconrec.2010.04.003>.
- Bulak, P., Polakowski, C., Nowak, K., Waśko, A., Wiącek, D., Bieganski, A., 2018. *Hermetia illucens* as a new and promising species for use in entomoremediation. *Sci. Total Environ.* 633, 912–919. <https://doi.org/10.1016/j.scitotenv.2018.03.252>.
- Cai, M., Ma, S., Hu, R., Tomberlin, J.K., Yu, C., Huang, Y., Zhan, S., Li, W., Zheng, L., Yu, Z., Zhang, J., 2018. Systematic characterization and proposed pathway of tetracycline degradation in solid waste treatment by *Hermetia illucens* with intestinal microbiota. *Environ. Pollut.* 242, 634–642. <https://doi.org/10.1016/j.envpol.2018.06.105>.
- Cencic, O., Rechberger, H., 2008. Material flow analysis with software STAN. *Environ. Inform. Sci.* 2008, pp. 440–447.
- Chen, J., Hou, D., Pang, W., Nowak, E.E., Tomberlin, J.K., Hu, R., Chen, H., Xie, J., Zhang, J., Yu, Z., Li, Q., 2019. Effect of moisture content on greenhouse gas and NH₃ emissions from pig manure converted by black soldier fly. *Sci. Total Environ.* 697, 133840. <https://doi.org/10.1016/j.scitotenv.2019.133840>.
- Chung, Y.C., 2007. Evaluation of gas removal and bacterial community diversity in a biofilter developed to treat composting exhaust gases. *J. Hazard. Mater.* 144, 377–385. <https://doi.org/10.1016/j.jhazmat.2006.10.04>.
- Clavreul, J., Guyonnet, D., Christensen, T.H., 2012. Quantifying uncertainty in LCA-modelling of waste management systems. *Waste Manag.* 32, 2482–2495. <https://doi.org/10.1016/j.wasman.2012.07.008>.
- Clavreul, J., Baumeister, H., Christensen, T.H., Damgaard, A., 2014. An environmental assessment system for environmental technologies. *Environ. Model. Softw.* 60, 18–30. <https://doi.org/10.1016/j.envsoft.2014.06.007>.
- Diener, S., Studt Solano, N.M., Roa Gutiérrez, F., Zurbrugg, C., Tockner, K., 2011. Biological treatment of municipal organic waste using black soldier fly larvae. *Waste Biomass Valorization* 2, 357–363. <https://doi.org/10.1007/s12649-011-9079-1>.
- Ecoinvent, 2007. Ecoinvent database [WWW document]. World's most consistent transparent life cycle inventory. Database. Switzerland. <https://www.ecoinvent.org/>.
- Ermolaev, E., Lalander, C., Vinnerås, B., 2019. Greenhouse gas emissions from small-scale fly larvae composting with *Hermetia illucens*. *Waste Manag.* 96, 65–74. <https://doi.org/10.1016/j.wasman.2019.07.011>.
- FAO, 2013. Edible insects. Future Prospects for Food and Feed Security, Future Prospects for Food and Feed Security. <https://doi.org/10.1017/CBO9781107415324.004>.
- FAO, 2018. Food Loss and Waste Reduction [WWW Document]. Food Agric. Organ.
- Gold, M., Tomberlin, J.K., Diener, S., Zurbrugg, C., Mathys, A., 2018. Decomposition of biowaste macronutrients, microbes, and chemicals in black soldier fly larval

- treatment: a review. *Waste Manag.* 82, 302–318. <https://doi.org/10.1016/j.wasman.2018.10.022>.
- Guo, H., Zhao, Y., Damgaard, A., Wang, Q., Lu, W., Wang, H., Christensen, T.H., 2019. Material flow analysis of alternative biorefinery systems for managing Chinese food waste. *Resour. Conserv. Recycl.* 149, 197–209. <https://doi.org/10.1016/j.resconrec.2019.05.010>.
- Hansen, T.L., Bhandar, G.S., Christensen, T.H., Bruun, S., Jensen, L.S., 2006. Life cycle modelling of environmental impacts of application of processed organic municipal solid waste on agricultural land (Easewaste). *Waste Manag. Res.* 24, 153–166. <https://doi.org/10.1177/0734242X06063053>.
- Hayes, A.C., Enongene Ekwe, S., Mervin, S., Jensen, E., 2016. Techno-economic evaluation of a tandem dry batch, garage-style digestion-compost process for remote work camp environments. *Waste Manag.* 58, 70–80. <https://doi.org/10.1016/j.wasman.2016.09.043>.
- Henry, M., Gasco, L., Piccolo, G., Fountoulaki, E., 2015. Review on the use of insects in the diet of farmed fish: past and future. *Anim. Feed Sci. Technol.* 203, 1–22. <https://doi.org/10.1016/j.anifeedsci.2015.03.001>.
- Hodge, K.L., Levis, J.W., DeCarolis, J.F., Barlaz, M.A., 2016. Systematic evaluation of industrial, commercial, and institutional food waste management strategies in the United States. *Environ. Sci. Technol.* 50, 8444–8452.
- Huailong, Y., Xiaoyan, H., 2018. Research on the resource utilization of black soldier fly (in Chinese). *Siliao Gongye*, 39. <https://doi.org/10.13302/j.cnki.fi.2018.06.013>.
- Huis, A. Van, Litterbeek, J. Van, Klunder, H., Mertens, E., Halloran, A., Muir, G., Vantomme, P., 2013. *EDIBLE INSECTS Future Prospects for Food and Feed Security*. Food and Agriculture Organization of the United Nations.
- Ites, S., Smetana, S., Toepfl, S., Heinz, V., 2020. Modularity of insect production and processing as a path to efficient and sustainable food waste treatment. *J. Clean. Prod.* 248. <https://doi.org/10.1016/j.jclepro.2019.119248>.
- Jensen, M.B., Möller, J., Scheutz, C., 2017. Assessment of a combined dry anaerobic digestion and post-composting treatment facility for source-separated organic household waste, using material and substance flow analysis and life cycle inventory. *Waste Manag.* 66, 23–35. <https://doi.org/10.1016/j.wasman.2017.03.029>.
- Jiang, C., 2018. *Effects of Gut Microbiome of Black Soldier Fly Larvae (Hermetia illucens) on the Biodegradation of Food Waste*. Master thesis. Zhejiang University (In Chinese).
- Jiang, C.L., Jin, W.Z., Tao, X.H., Zhang, Q., Zhu, J., Feng, S.Y., Xu, X.H., Li, H.Y., Wang, Z.H., Zhang, Z.J., 2019. Black soldier fly larvae (*Hermetia illucens*) strengthen the metabolic function of food waste biodegradation by gut microbiome. *Microb. Biotechnol.* 12, 528–543. <https://doi.org/10.1111/1751-7915.13393>.
- Khan, S.H., 2018. Recent advances in role of insects as alternative protein source in poultry nutrition. *J. Appl. Anim. Res.* 46, 1144–1157. <https://doi.org/10.1080/09712119.2018.1474743>.
- Klinglmaier, M., Zabol, O., Laner, D., Rechberger, H., Astrup, T.F., Scheutz, C., 2016. The effect of data structure and model choices on MFA results: a comparison of phosphorus balances for Denmark and Austria. *Resour. Conserv. Recycl.* 109, 166–175. <https://doi.org/10.1016/j.resconrec.2016.03.009>.
- Kumar, S., Negi, S., Mandpe, A., Singh, R.V., Hussain, A., 2018. Rapid composting techniques in Indian context and utilization of black soldier fly for enhanced decomposition of biodegradable wastes – a comprehensive review. *J. Environ. Manag.* 227, 189–199. <https://doi.org/10.1016/j.jenvman.2018.08.096>.
- Lalander, C.H., Fijdeland, J., Diener, S., Eriksson, S., Vinnerås, B., 2014. High waste-to-biomass conversion and efficient *Salmonella* spp. reduction using black soldier fly for waste recycling. *Agron. Sustain. Dev.* 35, 261–271. <https://doi.org/10.1007/s13593-014-0235-4>.
- Lalander, C., Nordberg, A., Vinnerås, B., 2018. A comparison in product-value potential in four treatment strategies for food waste and faeces – assessing composting, fly larvae composting and anaerobic digestion. *GCB Bioenergy* 10, 84–91. <https://doi.org/10.1111/gcbb.12470>.
- Lalander, C., Diener, S., Zurbrugg, C., Vinnerås, B., 2019. Effects of feedstock on larval development and process efficiency in waste treatment with black soldier fly (*Hermetia illucens*). *J. Clean. Prod.* 208, 211–219. <https://doi.org/10.1016/j.jclepro.2018.10.017>.
- Laner, D., Fekets, J., Rechberger, H., Fellner, J., 2016. A novel approach to characterize data uncertainty in material flow analysis and its application to plastics flows in Austria. *J. Ind. Ecol.* 20, 1050–1063. <https://doi.org/10.1111/jiec.12326>.
- Laurent, A., Clavreul, J., Bernstad, A., Bakas, I., Niero, M., Gentil, E., Christensen, T.H., Hauschild, M.Z., 2014. Review of LCA studies of solid waste management systems – part II: methodological guidance for a better practice. *Waste Manag.* 34, 589–606. <https://doi.org/10.1016/j.wasman.2013.12.004>.
- Li, Q., Zheng, L., Qiu, N., Cai, H., Tomberlin, J.K., Yu, Z., 2011. Bioconversion of dairy manure by black soldier fly (Diptera: Stratiomyidae) for biodiesel and sugar production. *Waste Manag.* 31, 1316–1320. <https://doi.org/10.1016/j.wasman.2011.01.005>.
- Li, Y., Jin, Y., Borrión, A., Li, H., Li, J., 2017. Effects of organic composition on the anaerobic biodegradability of food waste. *Bioresour. Technol.* 243, 836–845. <https://doi.org/10.1016/j.biortech.2017.07.028>.
- Li, W., Li, Q., Wang, Y.Y., Zheng, L., Zhang, Y., Yu, Z., Chen, H., Zhang, J., 2018. Efficient bio-conversion of organic wastes to value-added chemicals by soaking, black soldier fly (*Hermetia illucens* L.) and anaerobic fermentation. *J. Environ. Manag.* 227, 267–276. <https://doi.org/10.1016/j.jenvman.2018.08.084>.
- Lim, J.W., Mohd-Noor, S.N., Wong, C.Y., Lam, M.K., Goh, P.S., Beniers, J.J.A., Oh, W. Da, Jumbri, K., Ghani, N.A., 2019. Palatability of black soldier fly larvae in valorizing mixed waste coconut endosperm and soybean curd residue into larval lipid and protein sources. *J. Environ. Manag.* 231, 129–136. <https://doi.org/10.1016/j.jenvman.2018.10.022>.
- Liu, X., Chen, X., Wang, H., Yang, Q., Ur Rehman, K., Li, W., Cai, M., Li, Q., Mazza, L., Zhang, J., Yu, Z., Zheng, L., 2017. Dynamic changes of nutrient composition throughout the entire life cycle of black soldier fly. *PLoS One* 12. <https://doi.org/10.1371/journal.pone.0182601>.
- Liu, T., Awasthi, M.K., Awasthi, S.K., Duan, Y., Zhang, Z., 2020. Effects of black soldier fly larvae (Diptera: Stratiomyidae) on food waste and sewage sludge composting. *J. Environ. Manag.* 256, 109967. <https://doi.org/10.1016/j.jenvman.2019.109967>.
- Meneguz, M., Schiavone, A., Gai, F., Dama, A., Lussiana, C., Renna, M., Gasco, L., 2018. Effect of rearing substrate on growth performance, waste reduction efficiency and chemical composition of black soldier fly (*Hermetia illucens*) larvae. *J. Sci. Food Agric.* 98, 5776–5784. <https://doi.org/10.1002/jsfa.9127>.
- Mertenat, A., Diener, S., Zurbrugg, C., 2019. Black soldier Fly biowaste treatment – assessment of global warming potential. *Waste Manag.* 84, 173–181. <https://doi.org/10.1016/j.wasman.2018.11.040>.
- Naumann, C., Bassler, R., Seibold, R., Barth, C., 1997. *Methodenbuch: Die Chemische Untersuchung von Futtermitteln*. VdLUFA-Verlag.
- Newton, G.L., Sheppard, D.C., Watson, D., Burtle, G.J., Dove, C.R., Tomberlin, J.K., Thelen, E.E., 2005. The black soldier fly, *hermetia illucens*, as a manure management/resource recovery tool. *Symp. State Sci. Anim. Manure Waste Manag.* vol. 57, pp. 275–296.
- Pang, W., Hou, D., Chen, J., Nowar, E.E., Li, Z., Hu, R., Tomberlin, J.K., Yu, Z., Li, Q., Wang, S., 2020. Reducing greenhouse gas emissions and enhancing carbon and nitrogen conversion in food wastes by the black soldier fly. *J. Environ. Manag.* 260, 110066. <https://doi.org/10.1016/j.jenvman.2020.110066>.
- Peigné, J., Girardin, P., 2004. Environmental impacts of farm-scale composting practices. *Water Air Soil Pollut.* 153, 45–68. <https://doi.org/10.1023/B:WATE.0000019932.04020.b6>.
- Rehman, K., Rehman, R., Somroo, A., Cai, M., Zheng, L., Xiao, X., Rehman, A., Rehman, A., Tomberlin, J., Yu, Z., Zhang, J., 2017. Conversion of mixtures of dairy manure and soybean curd residue by black soldier fly larvae (*Hermetia illucens* L.). *J. Clean. Prod.* 154, 366–373. <https://doi.org/10.1016/j.jclepro.2017.04.019>.
- Ren, X., Wang, Q., Awasthi, M.K., Zhao, J., Wang, J., Liu, T., Li, R., Zhang, Z., 2019. Improvement of cleaner composting production by adding diatomite: from the nitrogen conservation and greenhouse gas emission. *Bioresour. Technol.* 286, 121377. <https://doi.org/10.1016/j.biortech.2019.121377>.
- Salomone, R., Saija, G., Mondello, G., Giannetto, A., Fasulo, S., Savastano, D., 2017. Environmental impact of food waste bioconversion by insects: application of life cycle assessment to process using *Hermetia illucens*. *J. Clean. Prod.* 140, 890–905. <https://doi.org/10.1016/j.jclepro.2016.06.154>.
- Shisheng, L., 2016. *The Study on Replacement of Fish Meal With Black Soldier Fly Meal in Feed for Common Carp (Cyprinus carpio)*. Northwest A & F University.
- Smetana, S., Palanisamy, M., Mathys, A., Heinz, V., 2016. Sustainability of insect use for feed and food: life cycle assessment perspective. *J. Clean. Prod.* 137, 741–751. <https://doi.org/10.1016/j.jclepro.2016.07.148>.
- Smetana, S., Schmitt, E., Mathys, A., 2019. Sustainable use of *Hermetia illucens* insect biomass for feed and food: Attributional and consequential life cycle assessment. *Resour. Conserv. Recycl.* 144, 285–296. <https://doi.org/10.1016/j.resconrec.2019.01.042>.
- Song, X., Shi, Z., Li, X., Wang, X., Ren, Y., 2019. Fate of proteins of waste activated sludge during thermal alkali pretreatment in terms of sludge protein recovery. *Front. Environ. Sci. Eng.* 13. <https://doi.org/10.1007/s11783-019-1114-7>.
- Starostina, V., Damgaard, A., Eriksen, M.K., Christensen, T.H., 2018. Waste management in the Irkutsk region, Siberia, Russia: an environmental assessment of alternative development scenarios. *Waste Manag. Res.* 36, 373–385. <https://doi.org/10.1177/0734242X18757627>.
- Steindl-Rast, D., Lebell, S., 1997. 1. Introduction. *Practical Handbook of MFA*.
- Surendra, K.C., Olivier, R., Tomberlin, J.K., Jha, R., Khanal, S.K., 2016. Bioconversion of organic wastes into biodiesel and animal feed via insect farming. *Renew. Energy* 98, 197–202. <https://doi.org/10.1016/j.renene.2016.03.022>.
- Wang, Y., Xue, W., Zhu, Z., Yang, J., Li, X., Tian, Z., Dong, H., 2019. Mitigating ammonia emissions from typical broiler and layer manure management – a system analysis. *Waste Manag.* 93, 23–33. <https://doi.org/10.1016/j.wasman.2019.05.019>.
- Win, S.S., Ebner, J.H., Brownell, S.A., Pagano, S.S., Cruz-Diloné, P., Trabold, T.A., 2018. Anaerobic digestion of black soldier fly larvae (BSFL) biomass as part of an integrated biorefinery. *Renew. Energy* 127, 705–712. <https://doi.org/10.1016/j.renene.2018.04.093>.
- Wong, C.Y., Rosli, S.S., Uemura, Y., Ho, Y.C., Leejeerajumnean, A., Kiatkittipong, W., Cheng, C.K., Lam, M.K., Lim, J.W., 2019. Potential protein and biodiesel sources from black soldier fly larvae: insights of larval harvesting instar and fermented feeding medium. *Energies* 12. <https://doi.org/10.3390/en12081570>.
- Wu, D., Feng, Y., Xue, L., Liu, M., Yang, B., Hu, F., Yang, L., 2019. Biochar combined with vermicompost increases crop production while reducing ammonia and nitrous oxide emissions from a paddy soil. *Pedosphere* 29, 82–94. [https://doi.org/10.1016/S1002-0160\(18\)60050-5](https://doi.org/10.1016/S1002-0160(18)60050-5).
- van Zanten, H.H.E., Mollenhorst, H., Oninck, D.G.A.B., Bikker, P., Meerburg, B.G., de Boer, I.J.M., 2015. From environmental nuisance to environmental opportunity: housefly larvae convert waste to livestock feed. *J. Clean. Prod.* 102, 362–369. <https://doi.org/10.1016/j.jclepro.2015.04.106>.
- Zhang, H., Wen, Z., Chen, Y., 2016. Environment and economic feasibility of municipal solid waste central sorting strategy: a case study in Beijing. *Front. Environ. Sci. Eng.* 10, 1–11. <https://doi.org/10.1007/s11783-016-0852-z>.
- Zhao, Y., Christensen, T.H., Lu, W., Wu, H., Wang, H., 2011. Environmental impact assessment of solid waste management in Beijing City, China. *Waste Manag.* 31, 793–799. <https://doi.org/10.1016/j.wasman.2010.11.007>.
- Zhao, Y., Lu, W., Damgaard, A., Zhang, Y., Wang, H., 2015. Assessment of co-composting of sludge and woodchips in the perspective of environmental impacts (EASETECH). *Waste Manag.* 42, 55–60. <https://doi.org/10.1016/j.wasman.2015.04.021>.
- Zheng, L., Li, Q., Zhang, J., Yu, Z., 2012. Double the biodiesel yield: rearing black soldier fly larvae, *Hermetia illucens*, on solid residual fraction of restaurant waste after grease extraction for biodiesel production. *Renew. Energy* 41, 75–79. <https://doi.org/10.1016/j.renene.2011.10.004>.